

A Multi-Metric Characterization of Quantum Convolutional Neural Networks: Expressibility, Trainability, and Noise Sensitivity

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ABSTRACT

Quantum convolutional neural networks (QCNNs) are structured variational quantum architectures for quantum machine learning (QML). This study examines their practical reliability through the balance between expressibility, trainability, optimization geometry, and noise resilience. We propose a multi-metric QCNN framework integrating Haar-based expressibility analysis, gradient sensitivity assessment, finite-size scaling, optimizer benchmarking, empirical Fisher information, Hessian spectral analysis, parameter efficiency evaluation, and depolarizing noise characterization. Our applied 6-qubit, 3-layer QCNN-inspired architecture, with 72 trainable parameters, achieved a Kolmogorov-Smirnov (KS) similarity of (0.8826 ± 0.0144) and a Kullback-Leibler (KL) divergence of (0.05648 ± 0.00578) relative to the Haar-random fidelity distribution. Increasing circuit depth improved expressibility, increasing KS similarity from 0.7400 to 0.9471 and reducing KL divergence from 0.1812 to 0.01815. However, 30 of 72 parameters exhibited near-zero sensitivity, and the gradient variance decreased from 0.13939 to 0.00929 across 2–8 qubits with no evidence of asymptotic barren plateaus (BP). Furthermore, Adam achieved lower mean squared error (MSE) than gradient descent (0.24388 vs. 0.44056), while Fisher and Hessian analyses revealed anisotropic optimization landscapes. The QCNN required 72 parameters compared with 906 for a dense neural network (DNN) baseline. Under depolarizing noise, MSE increased from 0.39506 to 0.63404 and purity decreased from 1.0 to 0.0893, demonstrating that expressibility alone does not ensure robust trainability.

Keywords: Quantum Convolutional Neural Network (QCNN), Expressibility, Trainability, Optimization, Depolarizing Noise.

Introduction

Machine learning (ML) has emerged as significant in extracting insightful interpretations from extensive and intricate datasets. In the era of classical learning approaches, convolutional neural networks (CNNs) performed quite well because they are proficient at handling structured data like images and time-series signals¹ by incorporating specific filters, parameter sharing, and hierarchical feature extraction. Their applications span computer vision (CV)², natural language processing (NLP)³, and many data-heavy endeavors.

With the escalating complexity of datasets and model designs, the computational needs of conventional ML systems remain a catalyst for the investigation of different computational paradigms. One such paradigm is quantum computing (QC), which has emerged and has the potential to offer novel solutions to computational problems that are still difficult to solve for classical systems^{4–6}. Combining quantum information with ML methods, QML has been studied for its use in learning tasks, computing tasks that use few resources, and representing quantum data^{7–9}. These strategies utilize quantum phenomena like superposition, interference, and entanglement¹⁰, which have driven the creation of quantum-enhanced learning models¹¹. Within this paradigm, QCNNs enhance convolutional processing for quantum systems by employing parameterized unitary operations, local entangling transformations, and pooling-type methods¹².

Quantum neural networks (QNNs) have gained popularity as variational architectures for QC and QML applications^{13–16}. QNN-based models have been studied in various fields, including quantum chemistry¹⁷, financial modeling¹⁸, and quantum sensing^{19,20}. Many such models rely on parameterized quantum circuits (PQCs) for their computational core, which provide trainable quantum transformations over high-dimensional Hilbert spaces. Several QNN architectures have since been developed, including quantum generative adversarial networks (QGANs)²¹, recurrent QNNs^{22,23}, convolutional QNNs^{12,24,25}, and

randomized quantum neural architectures²⁶.

The operational efficiency of these variational quantum methods is significantly influenced by two associated characteristics: expressibility and trainability. Expressibility refers to the capability of a parameterized circuit to yield a sufficiently diverse collection of quantum states, whereas trainability addresses the presence of informative gradients during the optimization of parameters²⁷. These attributes are fundamentally associated with circuit depth, connectivity, parameterization, and the topology of the optimization landscape.

A significant hurdle encountered in the training of QNNs is the barren-plateau phenomenon. This happens when gradient variances decline rapidly as the system scale expands, producing flatter optimization surfaces that complicate parameter modifications²⁸. In order to resolve these trainability concerns, other solutions have been recommended. This includes modified circuit couplings and branching frameworks²⁹, hybrid quantum-classical preprocessing and postprocessing methods^{30,31}, adaptive or randomized circuit architectures, and alternative parameterizations for QCNNs^{32,33}. At the same time, both theoretical and numerical research has scrutinized the expressibility of quantum circuits through the evaluation of measures pertaining to state distributions, entanglement, and the breadth of quantum-state space³⁴.

Despite these advancements, a comprehensive characterization of QCNN behavior requires evaluating expressibility and trainability together rather than separately. Current research has prompted further investigation into how circuit architecture affects representational capacity and optimization performance. In particular, it is essential to explore how the expressibility of QCNNs varies with circuit depth, how gradient information changes as the circuit width increases, how the sensitivity of optimization is distributed among individual parameters, and how the local optimization geometry behaves around combinations of trained parameters. Additionally, it is important to assess the performance of variational quantum models under non-ideal conditions.

This research investigates a 6-qubit QCNN employing a three-layer foundational structure alongside controlled variations in depth and width. This investigation contributes four primary findings. Initially, the expressibility of the QCNN is assessed using paired fidelity distributions in comparison to the analytical Haar-random fidelity distribution, employing complementary KS and KL metrics alongside a thorough circuit-depth investigation. Secondly, trainability is assessed by gradient norms, coordinate-specific gradient variance, parameter sensitivity, and finite-size scaling across various qubit configurations. Third, the optimization behavior and local parameter geometry are delineated via a systematic comparison of gradient descent and Adam, followed by empirical assessments of Fisher information and the Hessian spectrum. Lastly, the research conducts a comparative analysis of structural parameters against a dense neural network (DNN). It examines the susceptibility to simulated local depolarizing noise via task error, prediction deviation, gradient attenuation, and output-state purity.

The subsequent sections of the article are structured as outlined below. Section 2 examines related studies on QNNs, QCNNs, expressibility, trainability, and barren plateaus (BP). Section 3 delineates the computational structure and methodological approaches. Section 4 presents the quantitative findings. Section 5 delineates the findings and outlines avenues for subsequent investigation.

Related Works

Significant research has already been conducted on QCNNs and their potential applications in QML. This research has opened up new possibilities for future advancements in this field. Chen et al.³⁵ developed a QCNN model for image categorization. For that, they introduced probabilistic amplitude-based quantum encoding to convert classical data into quantum states, lowering dimensionality and making the switch easier. Finally, they showed that their applied QCNN fared better than standard techniques for classifying quantum images when tested on the MNIST handwritten digit dataset. To facilitate image classification more effectively, Long et al.³⁶ formulated the hybrid quantum-classical-quantum convolutional neural network (HQC-QCNN). This model integrates classical CNNs with quantum modules. The model uses a CNN to extract spatial features, a fixed quantum convolutional filter to map quantum features, and a trainable QNN to classify the features. When tested on MNIST, Fashion-MNIST, and MRI brain tumour datasets, HQC-QCNN demonstrated superior performance compared to classical CNNs and other hybrid models, including QCCNN, with an accuracy of up to 98.85%. Additionally, the model demonstrated resilience against finite sampling effects and quantum noise, with moderate-depth quantum circuits ensuring expressive and stable learning. In another study, Wen et al.³⁷ presented a new quantum residual neural network, called QResNet, which enhances expressivity by utilizing quantum circuits with residual connections. Their model is capable of producing a broader range of Fourier spectrum frequency components by augmenting the data-encoding and trainable blocks with more qubits. Experimental results show that residual encoding dramatically reduces the MSE of the regression task to 5.1×10^{-5} compared to conventional methods. Residual encoding enhances image classification (MNIST) accuracy by 7.7%, reaching 92.35% in testing and 92.85% in training.

López et al.³⁸ developed a novel quantum trainability and expressivity evaluation, called Q-TEE, a service-oriented method, to evaluate variational quantum circuits' (VQC) expressivity and trainability. The approach analyses 5-, 10-, and 15-node circuits to determine VQC efficiency on optimization issues like MaxCut. This paper evaluates these ansätze' expressiveness

and trainability using their expectation value distributions. On the 5-qubit graph, all ansätze had identical expressivity, but the single-layer RY circuit had a much broader dispersion, indicating higher expressivity but inferior trainability. With 10 or 15 qubits in the graph, expressivity discrepancies became more obvious. In a different study, Tangpanitanon et al.³⁹ investigated the expressibility and trainability of parameterized analogue quantum systems, prioritizing driven quantum Ising chains. The system achieves quantum supremacy in all phases, including thermalised and many-body localized (MBL) states, a crucial strength. The authors suggest sequential training for generative modeling challenges. The results reveal that thermalised phases' chaotic dynamics distract, but MBL phases stabilize the system, improving trainability. Using a hardware-efficient ansatz (HEA), Chen et al.⁴⁰ examined the correlation between expressibility and overfitting in quantum circuit learning (QCL). As quantum circuit depth increases, expressibility and generalization error saturate; automated regularization may prevent overfitting. The work used Probably Approximately Correct (PAC) learnability and Vapnik-Chervonenkis (VC) dimension analysis to show that unitary and localization in the ansatz limit model expressibility and complexity. These results support the idea that quantum circuits have an innate regularization mechanism, which makes hardware restrictions and QML performance easier to overcome. Monbroussou et al.⁴¹ analyzed the expressivity and trainability of VQCs, which are quantum circuits that preserve Hamming weight. It presented a quantum data loader that used reconfigurable beam splitter (RBS) and fermionic beam splitter (FBS) gates to encode amplitudes into subspaces with fixed Hamming weights. To determine the controllability of these circuits, the authors utilized the quantum fisher information matrix (QFIM) rank. They showed that the gradient variance is limited under specific conditions, which allows them to avoid BP. A hybrid quantum-classical neural network with deep residual learning, named Res-HQCNN, which integrates quantum and conventional neural architectures, was proposed by Liang et al.⁴² to address graph-structured quantum data. This model acquired sophisticated quantum knowledge rapidly by integrating the expressiveness of QNNs and the stability of residual learning. With a focus on the impact of various network depths and residual block topologies, they conducted a series of investigations to evaluate the performance of Res-HQCNN on graph-structured quantum data. As a consequence of its residual connections, Res-HQCNN far surpasses a conventional universal QNN in terms of performance when dealing with graph-structured data. By integrating graph information into the learning process, Res-HQCNN consistently surpasses the baseline model in both testing and training phases simultaneously. In an assessment of competing models utilizing line-structured data and a single hidden layer, Res-HQCNN demonstrated superior performance, especially with the incorporation of graph information ($\gamma = -0.5$).

Liao et al.⁴³ explored ways to improve the expressive power of quantum circuit-based QNNs for ML applications. The study suggests a sinusoidal function-friendly embedding for precise sinusoidal function characterization. With these methods, QNNs may better approximate complicated functions, especially sinusoidal ones, while using fewer quantum resources. However, traditional methods like QCL performed poorly compared to these novel approaches. Mjaaseth et al.⁴⁴ assessed the expressibility and entangling capabilities of PQC as a methodological concern. Statistical approaches were used to pick states from a class of quantum circuits to construct three descriptions for these features. A Python QML framework simplifies the development, training, and evaluation of PQC. The work trains 10,000 random Iris dataset circuits for categorization. They then use correlation analysis to compare the model's performance to the descriptor values. The results reveal that expressibility and entangling circuits perform better in categorization tasks. Particularly focusing on BP, Kashif et al.⁴⁵ studied how data encoding, ansatz expressibility, and qubit entanglement affected the trainability of hybrid quantum neural networks (HQNNs). The work showed through an empirical analytic method that backpropagation, marked by diminishing gradients and suboptimal optimization, is intensified by entanglement in quantum layers and increased circuit depth expressibility. The authors conclude that entanglement differentially affects performance in multi-class classification tasks based on their examination of amplitude and angle encoding systems. In their study, Chinzei et al.⁴⁶ investigated the manner in which QNNs achieve a balance between the efficacy of gradient measurement and expressivity. The study shows that expressive QNNs cost more to measure gradients. Using symmetrical quantum circuit topology, they devised the stabilizer-logical product ansatz (SLPA) for efficient gradient measurement. When Feff is 4, the SLPA measures gradients most efficiently. It reduces training time and data processing compared to parameter-shift approaches. By completely maximizing the dynamical Lie algebra (DLA) dimension, the SLPA ensures high expressivity and fast gradient estimation.

Panadero et al.⁴⁷ investigated the effects of data encoding techniques, network depth, and qubit entanglement on the overall performance of QNN by focusing on regression tasks. They used Fourier analysis to measure the network's output and found that data encoding limits maximal expressivity. After prior findings, the research reveals that entangling gates, especially at the last layer, improve the QNN's approximation capabilities compared to local qubit measurements. In a separate study, Aktar et al.⁴⁸ developed a graph neural network (GNN) based method to predict the expressibility of PQC, a measure for their capacity to capture various quantum states. The PQC is a graph in the GNN paradigm, with nodes representing inputs, gates, and outputs, and edges representing information flow. After training on 12,000 randomly selected PQCs, the model includes global factors like circuit depth and qubit count. It employs SAGEConv layers to retrieve local node data. The model minimizes expressibility estimation computing costs with a root-mean-squared error (RMSE) of 0.03 on random PQCs and 0.06 on IBM Qiskit's RealAmplitude circuits. In one more study, Shao et al.⁴⁹ designed the 2MC-OBPPP framework, a standard

method for assessing trainability, expressibility, and noise resilience in parameterized quantum circuits. It reduces overhead by checking noisy circuits with polynomial-time estimators, reducing "noise bottlenecks" and minimizing overhead. The study found that small-amplitude damping noise temporarily improves BP by improving gradient variance and training efficacy while decreasing expressibility. In a subsequent study, Nagies et al.⁵⁰ investigated QNN expressivity using SWAP test circuits and found that quadratic activation functions struggled to learn complex functions like parity checks beyond two dimensions. They suggest a product-layer generalized SWAP test circuit to activate functions with higher polynomials. With this version, the QNN can train for high-dimensional parity checks and similar tasks. Its potential for use in both quantum and classical learning is demonstrated by the fact that, despite noise, it obtained an 84% accuracy on a three-dimensional parity check using real quantum hardware. On synthetic and real-world datasets (IRIS, MNIST), the modified architecture performs better. Using frame potentials and the Kullback-Leibler (KL) divergence of fidelity distributions, Maronese et al.⁵¹ measured the expressibility of QNN designs and found their proximity to the Haar distribution. The numerical results demonstrate that for systems up to $n=10$ qubits, Clifford-enhanced matrix product states (CMPS) can attain near-Haar expressibility with $D_{KL} \approx 10^{-3}$ for the bond dimension $\chi=1$. When $n = 10$, fully-QNNs (fQNNs) require approximately 12 layers (~360 parameters) to achieve the same level of expressibility. These results show that classically simulable architectures can achieve excellent expressibility.

Methodology

This research carried out a simulation-based framework to investigate the distributional expressibility, finite-size trainability, optimization behavior, local parameter geometry, structural parameterization, and simulated noise sensitivity of a QCNN. The primary architecture consisted of a 3-layer parameterized circuit with a 6-qubit capacity, which was subject to controlled depth- and width-scaled variations for scaling analyses.

Computational Framework and Data Preparation

The numerical experiments were conducted in Python with the help of PennyLane (0.45.1). The *default.qubit* backend was used for ideal quantum-circuit simulations, while mixed-state simulations containing depolarizing noise used the *default.mixed*. A predetermined random-seed approach was employed to maintain reproducibility in circuit sampling, parameter initialization, and data segmentation. For numerical trials, a master seed of 20260730 was applied, and `random_state=42` was employed for dataset generation.

A synthetic dataset for binary classification, comprising six numerical attributes, was created with the *make_classification* function from scikit-learn. The dataset consisted of 100 observations, featuring three informative features, devoid of any superfluous features, and one cluster for each class. The number of input features was chosen to correspond with the six-qubit quantum register. The features were standardized using *StandardScaler*, and the categorical labels were assigned to

$$y \in \{-1, +1\} \quad (1)$$

where y denotes the binary class label, encoded as (-1) or (+1). The dataset was divided into 80 samples for training and 20 samples for testing. A fixed 20-sample training subset was employed for computationally intensive gradient analyzes, while all 80 training samples were retained for empirical Fisher estimation.

QCNN Architecture

The fundamental element of QCNN was made up of a 6-qubit, 3-layer parameterized quantum framework. Each layer was made up of 18 parameters dedicated to single-qubit rotations and six pooling parameters. This arrangement produced a cumulative sum of 72 parameterized gate angles spread among the three layers. This framework was used for fundamental analyses of expressibility, trainability, optimization, empirical Fisher, Hessian, and noise, while controlled variations of the identical circuit category were implemented to investigate depth and qubit scaling.

Data Encoding and Variational Circuit

Each standardized input feature was encoded into one qubit using R_Y angle encoding,

$$|\psi_{in}(\mathbf{x})\rangle = \bigotimes_{q=0}^5 R_Y(x_q) |0\rangle_q \quad (2)$$

Here, $\mathbf{x}$ is the input-feature vector, x_q is the feature encoded on the q -th qubit, and R_Y is a rotation around the Y-axis. The index q varies across the six qubits. Each qubit in each variational layer was subjected to trainable rotations of R_Y , R_Z , and R_X . Following the single-qubit manipulations, entangling and pooling operations were performed to allocate information throughout the quantum register.

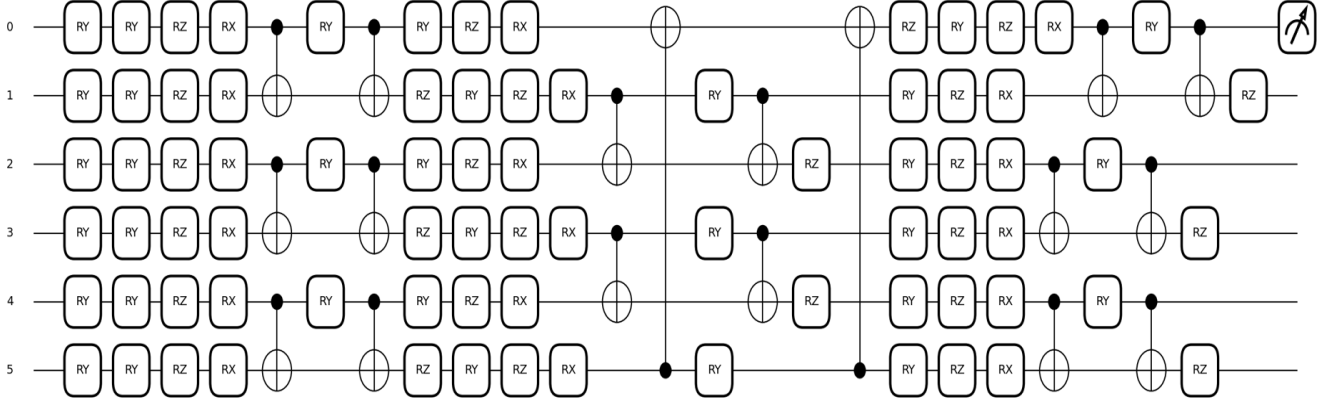


Figure 1. Gate-level representation of a 6-qubit QCNN architecture with parameterized rotation gates, CNOT entanglement, pooling-style operations, and Pauli-Z measurement.

Entanglement and Pooling-Style Operations

To ensure connectivity across all input qubits, an entanglement scheme featuring an alternating brick-wall pattern with periodic constraints was implemented. The qubit pairings were used in alternating layers within the six-qubit circuit.

$$(0, 1), (2, 3), (4, 5) \quad \text{and} \quad (1, 2), (3, 4), (5, 0) \quad (3)$$

The structured pairs define the interaction between the control qubits and target qubits applied in the alternate entangling layers. This connectivity schema facilitates the flow of information from all six qubits into the causal cone of the assessed output. A reverse causal-cone assessment affirmed that all six input qubits had the potential to influence the concluding Pauli-Z measurement. Furthermore, each associated subsystem had a trainable pooling-based transformation.

$$R_Y(\theta_c) \rightarrow \text{CNOT}(c, t) \rightarrow R_Z(\theta_t) \quad (4)$$

In this transformation, c and t represent the control and target qubits, respectively, while θ_c and θ_t indicate the relevant trainable rotation parameters. These actions are designated as pooling-style transformations since they provide pairwise trainable interactions without the physical extraction of qubits from the register.

Measurement and Cost Function

The circuit prediction was determined from the expectation value of the Pauli-Z operation on qubit 0,

$$\hat{y}(\mathbf{x}, \boldsymbol{\theta}) = \langle Z_0 \rangle \quad (5)$$

where $\hat{y}$ denotes the prediction QCNN, $\mathbf{x}$ denotes the input vector, $\boldsymbol{\theta}$ represents the vector of trainable parameters and $\langle Z_0 \rangle$ represents the expectation value Pauli-Z measured in qubit 0.

The model parameters were optimized using the mean squared error (MSE).

$$C(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N [\hat{y}_i - y_i]^2 \quad (6)$$

In Eq. 6, $C(\boldsymbol{\theta})$ denotes the MSE computed on the N samples, while $\hat{y}_i$ and y_i represent the predicted and true values corresponding to the i -th sample, respectively.

Expressibility Analysis

Fidelity-Based Expressibility

The evaluation of expressibility was conducted by comparing the distribution of pairwise fidelity from randomly parameterized QCNN states to the analytical fidelity distribution of Haar-random pure states. The definition of fidelity for two quantum states was established as:

$$F_{ij} = |\langle \psi_i | \psi_j \rangle|^2 \quad (7)$$

For a Hilbert space of dimension $d = 2^{N_q}$ corresponding to N_q qubits, the analytical Haar fidelity density is

$$P_{\text{Haar}}(F) = (d-1)(1-F)^{d-2} \quad (8)$$

In this instance, $P_{\text{Haar}}(F)$ denotes the Haar-random fidelity density associated with the fidelity parameter F .

Distributional similarity was quantified using the *Kolmogorov–Smirnov (KS)* distance and *Kullback–Leibler (KL)* divergence. In addition, a descriptive KS-based similarity measure was reported as

$$S_{\text{KS}} = 1 - D_{\text{KS}} \quad (9)$$

Participation-ratio metrics were further calculated as additional indications of sampled-state diversity employing a 200-state fidelity kernel in every iteration. The normalized participation ratio was calculated by dividing the sampled fidelity-kernel participation ratio by the number of kernel samples ($M=200$). The primary fidelity-distribution investigation included 10 independent iterations, featuring 250 randomly parameterized states and 2,000 sampled fidelity pairs for each iteration; KL divergence was assessed using 75 bins.

Circuit-Depth Analysis

To investigate the impact of circuit depth on distributional expressibility, identical circuit variety was assessed throughout one to six variational layers. Each depth was assessed in 10 separate iterations using 150 randomly configured states and 1,000 pairs of fidelity for each iteration. The resultant expressibility metrics were summarized with confidence intervals to delineate finite-depth patterns rather than to establish an optimal depth of the QCNN.

Trainability Analysis

Gradient and Parameter-Sensitivity Analysis

Trainability was characterized using gradients of the supervised objective with respect to the circuit parameters,

$$\mathbf{g} = \nabla_{\boldsymbol{\theta}} C(\boldsymbol{\theta}) \quad (10)$$

where $\mathbf{g}$ represents the gradient vector of the cost function associated with the vector of trainable parameters $\boldsymbol{\theta}$. The mean absolute parameter sensitivity, coordinate-wise gradient variance, and gradient norm were assessed across random parameter initializations to determine whether optimization information was uniformly distributed throughout the circuit or concentrated within a subset of parameter directions.

Parameter directions characterized by mean absolute gradient sensitivity not exceeding 10^{-12} were identified as numerically trivial. This tolerance was applied simply to recognize weakly influential parameter directions and was not viewed as a barren-plateau criterion. The first sensitivity assessment was carried out with 40 random parameter initializations based on the fixed analysis subset.

Finite-Size Gradient Scaling

Finite-size gradient scaling was studied for 2-, 4-, 6-, and 8-qubit circuits with a fixed depth of four layers. For each system size, five independent repeats were carried out, each with 20 random parameter initializations and a fixed 12-sample analysis subset. The approach was designed to describe the behavior of the finite-size gradient as the circuit width increased.

Optimization and Local Landscape Analysis

Optimization Procedure

Gradient descent and Adam were both initialized with the same parameter vector and optimized with the same objective function for 100 iterations at a learning rate of 0.01. Adam used

$$\beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon = 10^{-8} \quad (11)$$

In the context of the Adam optimizer, β_1 and β_2 indicate the decay rates for the first and second moments, respectively, and ϵ represents the constant that maintains numerical stability. For the following local landscape assessment, the Adam solution was optimized for up to 300 additional iterations utilizing a lowered learning rate of 0.003. A parameter vector was considered approximately stationary only when

$$\|\nabla C(\boldsymbol{\theta})\|_2 \leq 10^{-3} \quad (12)$$

The approximate stationarity was defined by the L_2 norm of the cost-function gradient, with a stationarity threshold set at 10^{-3} . The refinement phase was used exclusively for the local geometric assessment and was excluded from the direct optimizer evaluation.

Empirical Fisher Information

The empirical Fisher information matrix was used to characterize the geometry of the local parameters of the QCNN. For every training instance i , the gradient of the specific squared-error loss for the parameter vector $\boldsymbol{\theta}$ was calculated as

$$\mathbf{g}_i = \nabla_{\boldsymbol{\theta}} \ell_i(\boldsymbol{\theta}) \quad (13)$$

Here, ℓ_i indicates the distinct squared-error loss pertaining to the i -th sample. With all $N = 80$ training samples, the empirical Fisher matrix was computed as

$$\mathbf{F} = \frac{1}{N} \sum_{i=1}^N \mathbf{g}_i \mathbf{g}_i^T \quad (14)$$

The matrix $\mathbf{F}$ was created by computing the mean of the outer products $\mathbf{g}_i \mathbf{g}_i^T$ in the N training instances. The numerical rank, trace, eigenvalue spectrum, and conditioning of $\mathbf{F}$ were assessed to define the effective local dimensionality and anisotropy of the parameter space. Due to the potential rank deficiency of the empirical Fisher matrix, conditioning was evaluated across its nonzero eigenvalue spectrum and subsequent to diagonal regularization. The matrix that has been regularized was explicitly defined as

$$\mathbf{F}_{\text{reg}} = \mathbf{F} + 10^{-8} \mathbf{I} \quad (15)$$

Diagonal regularization was implemented using the identity matrix $\mathbf{I}$, utilizing 10^{-8} as the regularization coefficient. Subsequently, the condition number of $\mathbf{F}_{\text{reg}}$ was used as a numerically stabilized indicator of local conditioning.

Hessian Analysis

The Hessian of the cost function was utilized to analyze the second-order local curvature. The complete Hessian was computed using central finite differences of first-order gradients obtained by backpropagation.

$$H_{:,j} \approx \frac{\nabla C(\boldsymbol{\theta} + \varepsilon \mathbf{e}_j) - \nabla C(\boldsymbol{\theta} - \varepsilon \mathbf{e}_j)}{2\varepsilon} \quad (16)$$

In the finite-difference approximation, $\mathbf{e}_j$ is the unit vector along the j -th parameter direction, and ε is the size of the perturbation step. The finite-difference interval was determined to be $\varepsilon = 10^{-3}$, and the resulting matrix was symmetrically adjusted before eigendecomposition. The eigenvalues were identified as close to zero when

$$|\lambda_i| \leq \tau_H, \quad \tau_H = \max \left(10^{-10}, 10^{-6} \max_i |\lambda_i| \right) \quad (17)$$

where λ_i denotes the i -th eigenvalue of the Hessian matrix, and τ_H represents the threshold used to identify near-zero eigenvalues.

The Hessian spectrum was used to distinguish the positive, negative, and near-zero curvature directions. Critical-point labels like minimum, maximum, or saddle point were allocated only when the stationarity condition was met; if not, the spectrum was regarded as local curvature at a non-stationary assessment point.

Structural Parameter-Count Comparison

A fully connected dense neural network (DNN) with an architecture of [6, 32, 16, 8, 2] was evaluated for the structural parameter-count comparisons. The reference architecture involves an input layer with six features, three fully connected hidden layers with 32, 16, and 8 neurons, and an output layer with two neurons. For the hidden layers, Rectified Linear Unit (ReLU) activation was presumed, while the output activation was considered task-dependent.

The architecture comprises 848 weight parameters and 58 bias parameters, which add up to a total of 906 trainable parameters when biases are incorporated. This DNN was applied solely as a structural reference for parameter-count assessment. The relative model complexity was delineated in terms of the trainable parameter count by comparing its parameterization to the 72 trainable gate-angle parameters of the proposed QCNN.

Simulated Noise Analysis

Noise sensitivity was measured by applying a local depolarizing channel to each qubit individually after each QCNN layer.

$$\mathcal{E}_p(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad (18)$$

In this noise framework, ρ signifies the density matrix, p indicates the local depolarizing probability, and X , Y , and Z represent the Pauli operators. The local depolarization probability was

$$p \in \{0, 0.01, 0.05, 0.10\} \quad (19)$$

The enhanced QCNN parameter vector was maintained in several noise environments. Noise impacts were assessed using task MSE, deviation from perfect predictions, gradient norm, and purity of the output state.

$$\gamma(\rho) = \text{Tr}(\rho^2) \quad (20)$$

The purity of the state $\gamma(\rho)$ was calculated from the density matrix ρ , where Tr denotes the trace operation.

The MSE of the task, the prediction deviation, and gradient metrics were tested using a 20-sample test set. In contrast, output-state purity was estimated by utilizing five samples at each noise level. All evaluations were carried out using a controlled simulation based on a local depolarizing noise model.

Statistical Analysis

A summary of repeated numerical experiments was achieved using the arithmetic mean, sample standard deviation, and a two-sided 95% confidence interval.

$$\bar{x} \pm t_{0.975, n-1} \frac{s}{\sqrt{n}} \quad (21)$$

where $\bar{x}$ represents the mean of the sample, s signifies the standard deviation of the sample, and n indicates the count of independent trials. Here, we used confidence intervals to measure how uncertain things were in repeated experiments.

Results

This section provides a thorough multi-metric analysis of the associated QCNN, addressing distributional expressibility, finite-size trainability, optimization behavior, local parameter geometry, structural parameterization, and sensitivity to simulated noise. The study initially analyzes fidelity-based consensus with the Haar-random fidelity distribution and its reliance on circuit depth, followed by an exploration of parameter-level gradient sensitivity and qubit-scaling trends. Afterward, the local landscape is characterized by evaluating optimization performance alongside empirical Fisher and Hessian diagnostics. In the final studies, the trained QCNN's reaction to local depolarizing noise and the counts of structural parameters are examined. Altogether, these results yield a comprehensive delineation of the representational and optimization qualities of the applied QCNN architecture.

Expressibility of the Quantum Convolutional Neural Network

Table 1 presents an overview of the similarity in the distribution between the pairwise fidelity obtained by the 6-qubit, 3-layer QCNN and the analytical Haar-random fidelity distribution. Repeated sampling yielded an average KS distance of (0.1174 ± 0.0144) and a KS-based similarity of (0.8826 ± 0.0144) . The additive KL divergence was calculated to be (0.05648 ± 0.00578) . Both measures reveal that there is little run-to-run variance under the chosen sampling methodology, as indicated by the relatively narrow confidence intervals. The normalized participation ratio was (0.8709 ± 0.0031) , in contrast to the mean pairwise fidelity of (0.01568 ± 0.00019) . These findings show that the utilized QCNN creates a fidelity distribution that agrees with the Haar-random fidelity distribution in a large number of samples.

Table 1. Summary of distributional similarity and fidelity metrics.

Metric	Mean	SD	95% CI
KS-based similarity	0.8826	0.0144	[0.8724–0.8929]
KS distance	0.1174	0.0144	[0.1071–0.1276]
KL divergence	0.05648	0.00578	[0.05234–0.06061]
Sampled fidelity-kernel participation ratio	174.18	0.62	[173.73–174.62]
Normalized participation ratio	0.8709	0.0031	[0.8687–0.8731]
Mean pairwise fidelity	0.01568	0.00019	[0.01555–0.01582]

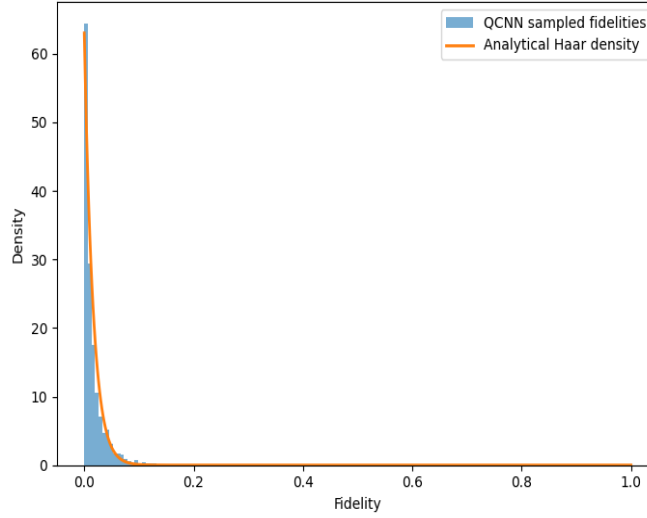


Figure 2. QCNN pairwise fidelity distribution relative to the Haar-random fidelity distribution.

Figure 2 illustrates the assessment of the distribution level. The QCNN fidelities obtained via sampling reproduce the primary low-fidelity concentration of the analytical Haar distribution, with barely discernible differences in the other fidelity ranges. The visual alignment is in accordance with the KS- and KL-focused data detailed in Table 1 and reinforces the interpretation of considerable distributional similarity.

Effect of Circuit Depth on Expressibility

A strong depth-dependent enhancement in expressibility was observed throughout the examined range (Table 2). The average KS-based similarity improved from (0.7400 ± 0.0153) at one layer to (0.9471 ± 0.0152) at six layers, whereas KL divergence diminished from (0.1812 ± 0.0363) to (0.01815 ± 0.00702) . The most significant change transpired between the one-layer and two-layer configurations, followed by increasingly minor enhancements at deeper levels. Significantly, both metrics exhibited a consistent shift toward more consistency with the Haar-random fidelity distribution, yielding convergent evidence that enhanced circuit depth augmented distributional expressibility, as measured by conformity with the Haar-random fidelity distribution, across the evaluated circuit structure.

Table 2. Depth-dependent KS similarity and KL divergence of the QCNN.

Layers	Parameters	KS similarity (mean $\pm$ SD)	95% CI	KL divergence (mean $\pm$ SD)	95% CI
1	24	0.7400 ± 0.0153	[0.7290–0.7509]	0.1812 ± 0.0363	[0.1552–0.2072]
2	48	0.8432 ± 0.0140	[0.8332–0.8533]	0.09483 ± 0.02240	[0.07881–0.11085]
3	72	0.8677 ± 0.0206	[0.8529–0.8824]	0.06364 ± 0.01429	[0.05342–0.07386]
4	96	0.8993 ± 0.0105	[0.8918–0.9068]	0.04226 ± 0.00709	[0.03719–0.04733]
5	120	0.9302 ± 0.0194	[0.9163–0.9441]	0.02737 ± 0.00881	[0.02107–0.03367]
6	144	0.9471 ± 0.0152	[0.9362–0.9579]	0.01815 ± 0.00702	[0.01313–0.02316]

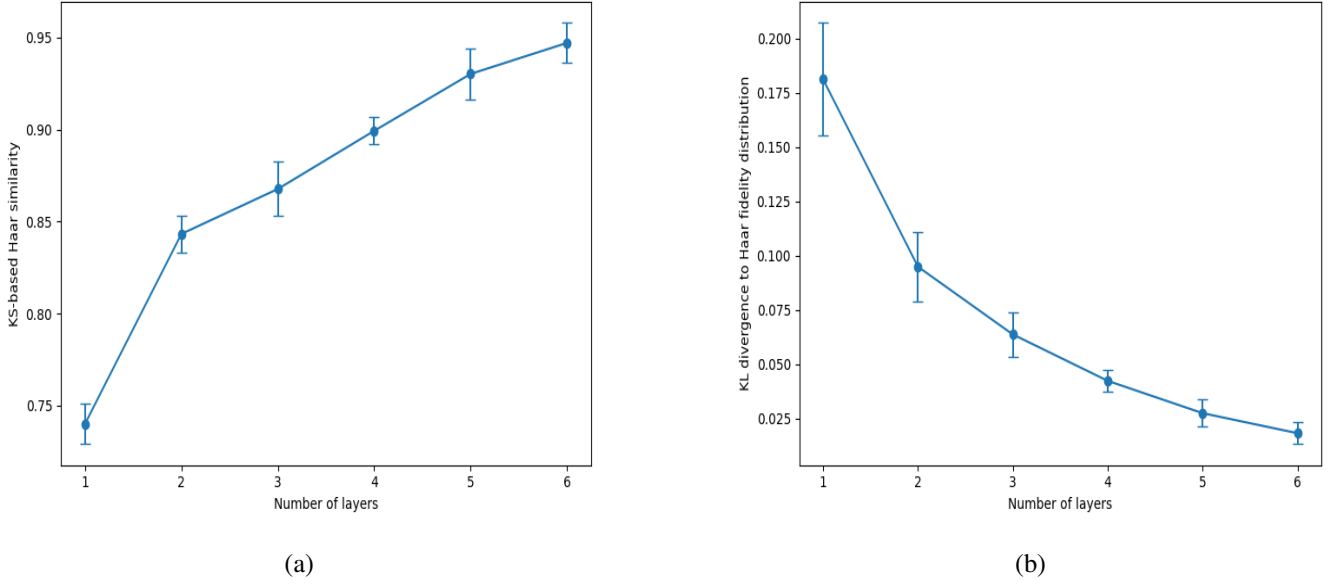


Figure 3. Depth dependence of QCNN expressibility: (a) KS-based similarity and (b) KL divergence.

The distinct trends are seen in Figure 3. The KS-based similarity exhibits a consistent increase with depth, as shown in Figure 3(a), but the KL divergence demonstrates a decline across the same interval, as shown in Figure 3(b). The 95% confidence intervals measure the variability across runs during repeated sampling while maintaining the overarching depth-dependent pattern. Finding that the two distributional measures are consistent reinforces the evidence that expressibility gets better as circuit depth increases, and neither tool is needed to find the best configuration on its own.

Finite-Size Trainability and Parameter Sensitivity

Table 3 provides a concise summary of the gradient and parameter-sensitivity characteristics of the applied QCNN. The mean gradient norm across random initializations was (1.1324 ± 0.3568) , suggesting a significant aggregate gradient signal. Nevertheless, a highly heterogeneous distribution of parameter sensitivities was revealed by the mean coordinate-wise gradient variance of 0.01943, which was much higher than the median of 0.000846. This trend was confirmed when 30 out of 72 parameterized gate angles matched the specified near-zero-sensitivity requirement. These findings show that optimization sensitivity is focused in several parameter directions, while many parameters are merely marginally important under the analyzed initialization regime. This variability at the parameter level does not prove the existence of a plateau on its own.

Table 3. Gradient statistics and sensitivity analysis.

Metric	Result
Mean gradient norm (mean $\pm$ SD)	(1.1324 ± 0.3568)
Mean coordinate-wise gradient variance	0.01943
Median coordinate-wise gradient variance	0.000846
Near-zero-sensitivity parameters	30/72
Near-zero-sensitivity fraction	41.67%

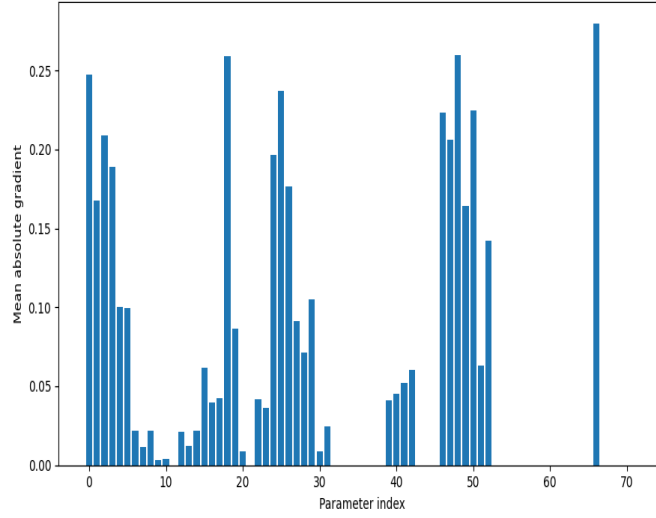


Figure 4. Parameter-wise gradient sensitivity across random initializations.

Figure 4 presents parameter-resolved sensitivities that further reveal this heterogeneity. Parameter intervals with modest or practically minimal mean absolute gradients are scattered within high-gradient areas. Disconnected input subsystems cannot simply explain these low-sensitivity orientations because the newly developed circuit layout connects all six input qubits to the measured causal cone. Rather, they are compatible with the unequal local influence of various circuit actions on the measured observable and parameter redundancy. The rank-deficient Fisher geometry and near-zero Hessian directions that are subsequently reported are significantly contextualized by this parameter-level structure.

Gradient-Variance Scaling with Qubit Count

As seen in Table 4, the coordinate-wise gradient variance decreases noticeably as the circuit width increases in the finite-size scaling experiment. At a constant depth of four layers, the mean variance diminished from 0.13939 for two qubits to 0.01934 for four qubits and 0.00927 for six qubits. Augmenting the system to eight qubits yielded minimal additional variation of 0.00929, exhibiting significant similarities between the confidence intervals of six and eight qubits. Consequently, the majority of the noted decrease transpires between two and six qubits, beyond which the finite-size trend remains rather constant within the anticipated margin of error. These findings indicate a gradient attenuation that varies with width throughout the evaluated systems. However, they do not confirm asymptotic exponential concentration nor, on their own, serve as assurance of a barren plateau.

Table 4. Mean coordinate-wise gradient variance across different qubit configurations.

Qubits	Layers	Parameters	Mean coordinate gradient variance	SD	95% CI
2	4	32	0.13939	0.08380	[0.03534 – 0.24345]
4	4	64	0.01934	0.00654	[0.01121 – 0.02746]
6	4	96	0.00927	0.00259	[0.00605 – 0.01248]
8	4	128	0.00929	0.00429	[0.00396 – 0.01462]

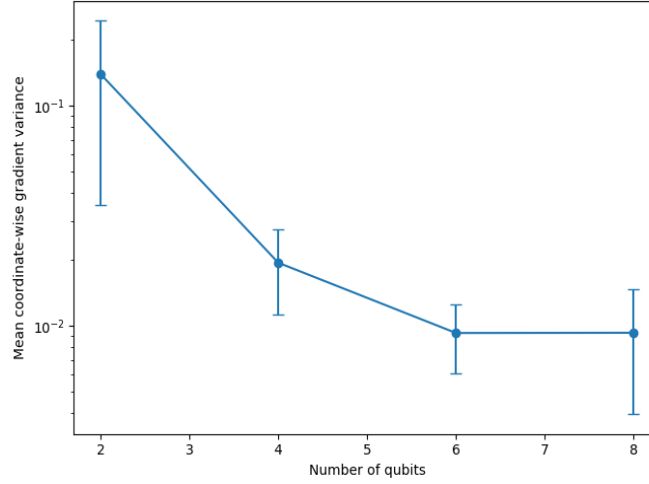


Figure 5. Finite-size gradient-variance scaling with qubit count.

Figure 5 illustrates the identical pattern on a logarithmic variance scale. The mean variance stays about the same at eight qubits, after a steep drop between two and four qubits and then a smaller drop at six qubits.

Optimization Behaviour

Under uniform initialization, both gradient descent and Adam began with an MSE of 1.44456 (Table 5). With the hyperparameter settings evaluated, Adam achieved a terminal training cost of 0.24388 after 100 iterations, whereas gradient descent decreased the objective to 0.44056. The variation manifested itself early in the training and continued throughout the optimization time frame. These results reveal that Adam facilitated a more potent empirical optimization for the current QCNN architecture, while the comparison is restricted to the selected initialization, learning rates, and finite iteration limits and should not be construed as a comprehensive optimization evaluation.

Table 5. Comparison of optimizer performance.

Optimizer	Initial MSE	Final MSE
Gradient Descent	1.44456	0.44056
Adam	1.44456	0.24388

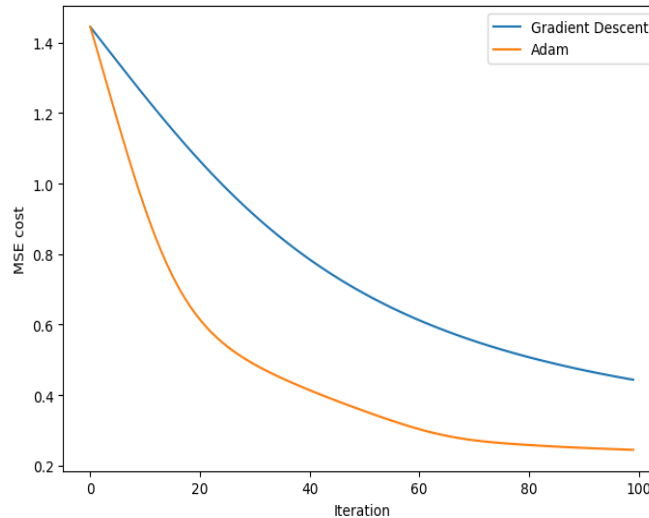


Figure 6. Optimization trajectories of gradient descent and Adam.

The optimization processes shown in Figure 6 elucidate the chronological patterns related to terminal expenses. Adam reveals a notably steeper initial drop in MSE and arrives at a lower-cost region more quickly than gradient descent, which adheres to a slower and more consistent descent. Both trajectories continue to decline during the 100-iteration interval, implying that the final values provided signify finite-budget optimization results rather than actual asymptotic convergence.

Local Fisher and Hessian Geometry

Table 6 presents a summary of the local optimization, empirical Fisher, and Hessian diagnostics at the improved Adam evaluation point. After further refinement with Adam, the desired value was reduced to 0.17119, and the gradient norm was achieved 0.00957. As this value persisted over the specified stationarity criterion of 0.001, the analyzed parameter vector was considered non-stationary, and no categorization of minimum, maximum, or saddle point was applied. The numerical rank of the empirical Fisher matrix was 25/72, suggesting that locally identified gradient information represented only 34.72% of the expected parameter space. The condition number of the Fisher nonzero spectrum, measured at 2.28×10^9 , in conjunction with the Fisher regularized condition number of 2.50×10^8 , reveals significant spectral anisotropy and substantial unevenness in sensitivity across parameter directions. The Hessian featured 30 positive, 7 negative, and 35 almost zero eigenvalues, with 48.61% of the spectrum identified as near zero. The eigenvalues spanned from -0.01095 to 3.7480 , reflecting a heterogeneous local curvature and a considerable fraction of weak-curvature trajectories. The combined analysis of the Fisher and Hessian diagnostics reveals a pronounced anisotropic local optimization landscape defined by a restricted set of impactful directions and a multitude of feebly responsive directions.

Table 6. Local optimization, empirical Fisher, and Hessian diagnostics.

Measure	Result
Refined Adam cost	0.17119
Gradient norm at refined point	0.00957
Stationarity threshold	0.001
Stationarity reached	No
Empirical Fisher numerical rank	25/72
Fisher nonzero-spectrum condition number	2.28×10^9
Fisher regularized condition number	2.50×10^8
Hessian positive eigenvalues	30
Hessian negative eigenvalues	7
Hessian near-zero eigenvalues	35
Near-zero Hessian fraction	48.61%
Hessian eigenvalue range	$[-0.01095, 3.7480]$

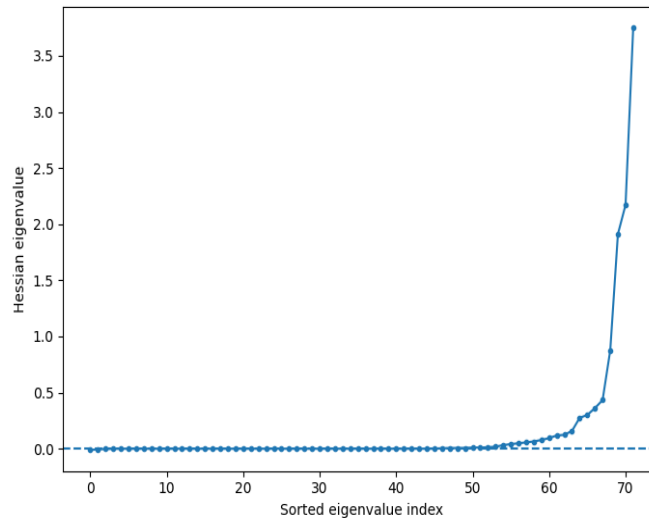


Figure 7. Hessian eigenvalue spectrum at the refined Adam evaluation point.

The highly skewed nature of the Hessian spectrum is demonstrated in Figure 7. A relatively modest positive tail explains the largest-curvature directions, whereas the majority of eigenvalues are concentrated near zero. However, the faint negative portion remains pertinent for the establishment of a mixed local curvature, despite being compressed in relation to the dominant positive eigenvalues that are present. This spectrum is consistent with the reduced empirical Fisher rank and the parameter-sensitivity analysis, both of which point to a local landscape featuring numerous weakly responsive or redundant parameter combinations and few influential directions.

Trainable-Parameter Comparison of QCNN and Dense Neural Network (DNN)

The fully connected DNN consists of 848 weights and 906 trainable coefficients, with 58 biases included, as shown in Table 7, whereas the simulated QCNN has 72 parameterized gate angles. As a result, the weight-only and bias-inclusive DNNs have approximately 11.8 and 12.6 times as many trainable coefficients as the QCNN, respectively. This analysis reveals a significant disparity in the size of parameterization among the chosen architectures.

Table 7. Structural trainable-parameter comparison between QCNN and DNN architectures.

Architecture	Parameter type	Trainable parameters
QCNN	Parameterized gate angles	72
DNN	Weights only	848
DNN	Weights + biases	906

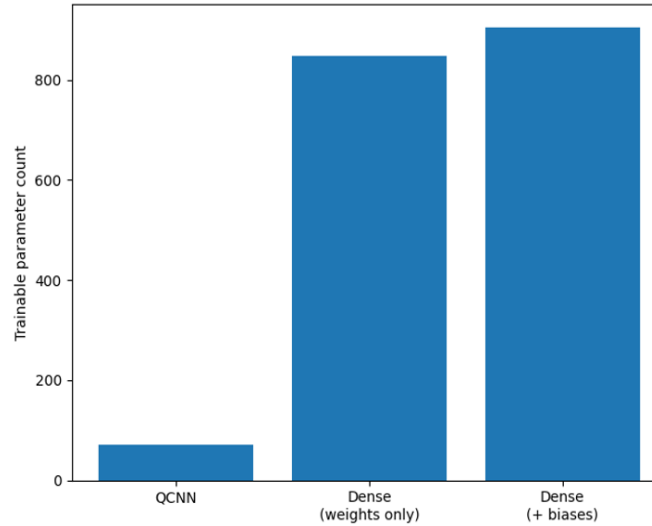


Figure 8. Structural parameter-count comparison of the QCNN and DNN.

The significance of this architectural difference is illustrated in Figure 8, which shows that the QCNN parameter count is significantly lower than both variants of the DNN.

Sensitivity to Simulated Depolarizing Noise

The simulated-noise experiment indicates an incremental reduction in task performance and gradient size as the local depolarization probability per QCNN layer increases (Table 8). Compared to the quiet MSE of 0.39506, the error increased to 0.41749 at $p=0.01$, 0.51393 at $p=0.05$, and 0.63404 at $p=0.10$. The prediction mean absolute error (MAE) relative to the ideal circuit increased from zero to 0.23790, as the gradient norm decreased from 0.55625 to 0.29588. At $p=0.10$, the gradient norm held just 53.19% of its noiseless value. These associated trends demonstrate that rising depolarizing noise adversely affects the task goal and lessens the optimization signal.

Table 8. QCNN performance and state purity under simulated local depolarizing noise.

Depolarizing probability (p)	MSE	Prediction MAE vs Ideal	Gradient norm	Relative gradient norm	Mean purity	Purity SD
0.00	0.39506	0.00000	0.55625	1.0000	1.0000	2.72×10^{-16}
0.01	0.41749	0.03028	0.52362	0.9413	0.7498	0.01867
0.05	0.51393	0.13595	0.41001	0.7371	0.2596	0.02955
0.10	0.63404	0.23790	0.29588	0.5319	0.0893	0.01604

Noise-related degradation was more marked at the state level. The mean purity reduced from 1.0 in the noiseless simulation to 0.7498, 0.2596, and 0.0893 for $p=0.01$, 0.05, and 0.10, respectively. The continuous decrease in purity denotes an intensification of state mixing as local depolarizing channels gather across the circuit. The slight shift in task mean squared error (MSE) at $p=0.01$, even with the drop in purity, implies that degradation at the state level does not directly impact immediate task-level error. At higher measured noise probabilities, the combined increase in MSE, the increase in prediction deviation, the decrease in gradient magnitude, and the reduction in purity furnish consistent proof of pronounced noise sensitivity.

The findings demonstrate the simultaneous presence of robust distributional expressibility alongside varied local trainability in the investigated QCNN. The consistency with the Haar-random fidelity distribution improved as the circuit depth increased. The fundamental 6-qubit design kept useful optimization signals, and Adam reached a lower training outcome with limited resources than gradient descent. At the parameter level, the gradient-sensitivity distribution, rank-deficient empirical Fisher geometry, and Hessian spectrum consistently revealed significant anisotropy and multiple weakly influencing directions. The finite-scale width-scaling experiment, moreover, demonstrated gradient attenuation with the enlargement of the system size; nevertheless, the trend discovered was inadequate to confirm asymptotic barren-plateau behavior. Similarly, simulated local depolarizing noise incrementally increased task errors, diminished gradient magnitude, and lowered output-state purity. These results collectively depict the employed QCNN as demonstrating significant distributional alignment with the Haar-random fidelity distribution at the examined scale, while underscoring the importance of weakly impactful parameter directions, local geometric anisotropy, finite-size gradient suppression, and noise sensitivity as critical limitations on trainability.

Conclusion

This research introduces an extensive multi-metric approach for assessing the intricate dynamics of QCNN-inspired variational quantum structures by concurrently examining expressibility, trainability, optimization geometry, parameter efficiency, and noise susceptibility. A connected 6-qubit, 3-layer QCNN-inspired architecture with 72 trainable parameters was systematically analyzed using complementary evaluation modules, including Haar-based fidelity expressibility assessment, gradient sensitivity analysis, finite-size scaling, optimizer comparison, empirical Fisher information, Hessian spectral analysis, structural parameter comparison, and simulated depolarizing noise evaluation. The findings show that better optimization features are not always correlated with higher representational capacity. Despite the circuit demonstrating significant similarity to Haar-random fidelity statistics and enhanced expressibility as depth increased, the analysis of trainability revealed considerable heterogeneity at the parameter level, with numerous slightly influential directions occupying a substantial portion of the parameter space. Moreover, Fisher and Hessian evaluations validated a considerably anisotropic optimization terrain distinguished by diminished effective dimensionality and broad low-curvature areas. However, the investigated range was insufficient to establish asymptotic barren-plateau behavior, as evidenced by the finite-size scaling results, which demonstrated gradient attenuation as circuit width increased. When considering optimization, Adam-based adaptive optimization had lower finite-budget error than gradient descent. However, noise studies showed that learning performance and quantum-state purity became worse as depolarization increased. The investigation of parameter efficiency further emphasized the concise parameterization of the QCNN in contrast to a fully connected classical baseline, while merely reducing parameters does not guarantee enhanced predictive performance.

This research is constrained by the diminutive simulated QCNN framework, synthetic dataset, and simplistic noise model, which may not adequately reflect extensive quantum systems or real-world hardware scenarios. Future investigations should broaden the suggested framework to include more extensive quantum circuits, real datasets, hardware-specific noise models, and sophisticated optimization techniques to assess the scalability, resilience, and practical applicability of QCNN-based learning architectures.

AUTHOR CONTRIBUTIONS STATEMENT

Methodology, T.H., A.A.R.; Software, T.H. and A.A.R.; Validation, T.H., A.A.R.; Formal analysis, T.H. and A.A.R.; Investigation, A.A.R., T.H.; Resources, A.A.R., T.H.; Writing - original draft, T.H., A.A.R.; Writing - review & editing, A.A.R., T.H.,

M.S.U. and O.; Visualization, T.H, A.A.R.; Supervision, O.; All authors have read and agreed to the published version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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RESOURCE AVAILABILITY

Data and code associated with this study will be made available by the corresponding author upon reasonable request to (tuhin97.hossain@gmail.com).

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